

## 5.5 DATA DICTIONARY — USER MODULE

FinanceOS: Investor & Personal Financial Management Schemas

USER SCOPE

### 1. User

Collection: users

Stores registered user identity, personal demographics, security credentials, contact details, and account standing.

| Field Name   | Data Type | Key    | Description                                        | Example                              |
|--------------|-----------|--------|----------------------------------------------------|--------------------------------------|
| _id          | ObjectId  | PK     | Primary key auto-generated by MongoDB              | ObjectId("65b8f1d2e13a4c001f8d901a") |
| userId       | String    | Unique | Unique customer identifier code                    | "USR-2026-0042"                      |
| name         | String    | —      | Full legal name of user                            | "Dip Jivrajani"                      |
| dateOfBirth  | Date      | —      | Date of birth for lifecycle profiling              | 2003-05-10                           |
| gender       | String    | —      | Gender identity of user                            | "Male"                               |
| phone        | String    | —      | Contact mobile phone number                        | "9876543210"                         |
| city         | String    | —      | Residential city of user                           | "Bhavnagar"                          |
| state        | String    | —      | Residential state                                  | "Gujarat"                            |
| email        | String    | Unique | Primary email address for login & notifications    | "dip@example.com"                    |
| role         | String    | —      | System access role: "user"   "Investor"            | "user"                               |
| status       | String    | —      | User account standing: Active, Inactive, Suspended | "Active"                             |
| otp          | String    | —      | Temporary one-time password for authentication     | "482910"                             |
| otpExpiresAt | Date      | —      | Expiration timestamp for issued OTP                | 2026-09-06T14:45:00.000Z             |
| createdAt    | Date      | —      | User account creation timestamp                    | 2026-08-08                           |
| updatedAt    | Date      | —      | Last profile update timestamp                      | 2026-08-08                           |

### 2. MonthlyFinance

Collection: monthlyfinances

Manages user monthly budget position, cash reserves, primary income, living expenses, and recurring commitments.

| Field Name      | Data Type | Key             | Description                                                    | Example                              |
|-----------------|-----------|-----------------|----------------------------------------------------------------|--------------------------------------|
| _id             | ObjectId  | PK              | Record identifier auto-generated by MongoDB                    | ObjectId("663a89bc2f1a900018a12345") |
| user            | ObjectId  | FK ,<br>Unique* | Reference to User document. *Compound unique with year + month | ObjectId("65b8f1d2e13a4c001f8d901a") |
| month           | Number    | Unique*         | Financial month of accounting period (1 to 12)                 | 7                                    |
| year            | Number    | Unique*         | Financial year of accounting period (2000 to 2100)             | 2026                                 |
| income          | Number    | —               | Total monthly earned primary income                            | 50000                                |
| expenses        | Number    | —               | Total monthly essential household and living expenses          | 32000                                |
| goalAllocations | Number    | —               | Total savings allocated toward active goals this period        | 10000                                |

| Field Name          | Data Type | Key     | Description                                            | Example    |
|---------------------|-----------|---------|--------------------------------------------------------|------------|
| cashBalance         | Number    | —       | Existing cash and savings balance (opening balance)    | 4000       |
| openingBalance      | Number    | —       | Carried-forward opening liquid cash balance            | 4000       |
| closingBalance      | Number    | —       | Projected or finalized month-end closing cash balance  | 22000      |
| commitments         | Number    | —       | Total monthly commitments (SIPs, premiums, loan EMI)   | 15000      |
| updateDate          | Date      | —       | Monthly finance update timestamp                       | 2026-07-05 |
| reminderEnabled     | Boolean   | —       | Monthly update reminder toggle                         | true       |
| emailNotification   | Boolean   | —       | Email reminder channel switch                          | true       |
| monthlySavings      | Number    | Derived | Net monthly surplus: (income - expenses)               | 18000      |
| availableToAllocate | Number    | Derived | Surplus available after commitments & goal allocations | 3000       |
| createdAt           | Date      | —       | Record creation timestamp                              | 2026-07-31 |
| updatedAt           | Date      | —       | Last modification timestamp                            | 2026-07-31 |

### 3. AdditionalIncome

Collection: additionalincomes

Tracks supplementary user inflows such as performance bonuses, freelancing, rental income, dividends, and interest payouts.

| Field Name   | Data Type     | Key | Description                                                       | Example                              |
|--------------|---------------|-----|-------------------------------------------------------------------|--------------------------------------|
| _id          | ObjectId      | PK  | Primary key of the income entry                                   | ObjectId("664b12345e6f789012345678") |
| user         | ObjectId      | FK  | Reference to User document receiving income                       | ObjectId("65b8f1d2e13a4c001f8d901a") |
| title        | String        | —   | Title or origin label of supplemental receipt                     | "Freelance UI Design"                |
| category     | Enum / String | —   | Classification: Bonus, Freelancing, Dividend, Rental, Gift, Other | "Freelancing"                        |
| amount       | Number        | —   | Monetary sum credited                                             | 25000                                |
| description  | String        | —   | Optional remarks regarding the receipt                            | "Website redesign milestone payout"  |
| month        | Number        | —   | Financial accounting month (1 to 12)                              | 7                                    |
| year         | Number        | —   | Financial accounting year                                         | 2026                                 |
| receivedDate | Date          | —   | Date when the funds were credited to account                      | 2026-07-15                           |
| createdAt    | Date          | —   | Record creation timestamp                                         | 2026-07-15                           |
| updatedAt    | Date          | —   | Record modification timestamp                                     | 2026-07-15                           |

### 4. SavingGoal

Collection: savinggoals

Tracks short-term and long-term financial target goals, accumulated balances, contribution frequencies, depository locations, and withdrawals.

| Field Name | Data Type | Key | Description                                   | Example                              |
|------------|-----------|-----|-----------------------------------------------|--------------------------------------|
| _id        | ObjectId  | PK  | Goal primary key identifier                   | ObjectId("665c23456a7b890123456789") |
| user       | ObjectId  | FK  | Reference to User document who owns this goal | ObjectId("65b8f1d2e13a4c001f8d901a") |

| Field Name               | Data Type         | Key         | Description                                                    | Example                          |
|--------------------------|-------------------|-------------|----------------------------------------------------------------|----------------------------------|
| goalName                 | String            | —           | Descriptive name of the saving goal                            | "Phone"                          |
| category                 | String            | —           | Goal category: Emergency Fund, Mobile, Laptop, Vehicle, Travel | "Mobile"                         |
| targetAmount             | Number            | —           | Target monetary amount required to achieve goal                | 100000                           |
| currentAmount            | Number            | —           | Current total accumulated savings balance                      | 25000                            |
| monthlyContribution      | Number            | —           | Planned monthly recurring savings contribution                 | 5000                             |
| contributionDay          | Number            | —           | Day of month (1-31) when contribution is due                   | 5                                |
| startDate                | Date              | —           | Commencement date of saving goal                               | 2026-01-01                       |
| targetDate               | Date              | —           | Target deadline date for achieving goal                        | 2027-02-08                       |
| fundLocation.type        | String            | —           | Asset holding account type (Bank, App, Cash)                   | "Bank Account"                   |
| fundLocation.institution | String            | Conditional | Depository bank or app name (if bank/app selected)             | "HDFC Bank"                      |
| fundLocation.lastFour    | String            | Conditional | Last 4 digits of depository account                            | "4821"                           |
| status                   | String            | —           | Goal standing: Active, Paused, Completed, Closed               | "Active"                         |
| contributions            | Array<br><Subdoc> | —           | Embedded list of deposit contributions (amount, date, source)  | [{ amount: 5000, date: ... }]    |
| withdrawals              | Array<br><Subdoc> | —           | Embedded list of emergency fund withdrawals                    | [{ amount: 2000, purpose: ... }] |
| goalProgress             | Number            | Derived     | Progress percentage toward completion (0 to 100)               | 25                               |
| createdAt                | Date              | —           | Goal creation timestamp                                        | 2026-08-08                       |
| updatedAt                | Date              | —           | Goal last update timestamp                                     | 2026-08-08                       |

## 5. Investment

Collection: investments

Manages user investment assets across Systematic Investment Plans (SIP), Fixed Deposits (FD), Recurring Deposits (RD), Gold, Mutual Funds, and Stocks.

| Field Name          | Data Type | Key | Description                                                   | Example                              |
|---------------------|-----------|-----|---------------------------------------------------------------|--------------------------------------|
| _id                 | ObjectId  | PK  | Investment asset identifier                                   | ObjectId("666d34567b8c901234567890") |
| user                | ObjectId  | FK  | Reference to User who holds the investment                    | ObjectId("65b8f1d2e13a4c001f8d901a") |
| type                | String    | —   | Investment category: "SIP", "Fixed Deposit", "Gold", "Stocks" | "Fixed Deposit"                      |
| name                | String    | —   | Name of investment product or fund                            | "HDFC Bank FD"                       |
| principalAmount     | Number    | —   | Base principal invested sum                                   | 500000                               |
| currentValue        | Number    | —   | Current evaluated market valuation of the asset               | 500000                               |
| monthlyContribution | Number    | —   | Recurring installment contribution amount (for SIP/RD)        | 5000                                 |
| interestRate        | Number    | —   | Annual percentage return / interest rate (p.a.)               | 6.25                                 |

| Field Name           | Data Type | Key         | Description                                      | Example     |
|----------------------|-----------|-------------|--------------------------------------------------|-------------|
| quantity             | Number    | —           | Number of units, shares, or grams of gold held   | 10          |
| purchasePrice        | Number    | —           | Acquisition price per unit / gram / share        | 500         |
| frequency            | String    | —           | Contribution interval (e.g. Monthly, Quarterly)  | "Monthly"   |
| startDate            | Date      | —           | Investment start / deposit date                  | 2026-01-01  |
| nextContributionDate | Date      | —           | Next scheduled installment due date              | 2026-09-01  |
| maturityDate         | Date      | —           | Maturity or lock-in expiration date              | 2027-01-01  |
| institution          | String    | Conditional | Bank, broker, AMC, or platform holding the asset | "HDFC Bank" |
| status               | String    | —           | Operational status: Active, Matured, Closed      | "Active"    |
| createdAt            | Date      | —           | Record creation timestamp                        | 2026-08-08  |
| updatedAt            | Date      | —           | Record modification timestamp                    | 2026-08-08  |

## 6. InsurancePolicy

Collection: insurances

Manages user risk protection policies covering Life, Health, Vehicle, and Home insurance, premium schedules, and renewals.

| Field Name       | Data Type | Key         | Description                                        | Example                              |
|------------------|-----------|-------------|----------------------------------------------------|--------------------------------------|
| _id              | ObjectId  | PK          | Insurance policy identifier                        | ObjectId("668f56789d0e123456789012") |
| user             | ObjectId  | FK          | Reference to policyholder User document            | ObjectId("65b8f1d2e13a4c001f8d901a") |
| type             | String    | —           | Insurance category: Life, Health, Vehicle, Home    | "Life Insurance"                     |
| name             | String    | —           | Policy commercial product name                     | "LIC Jeevan Labh"                    |
| provider         | String    | —           | Underwriting insurance corporation                 | "LIC of India"                       |
| policyNumber     | String    | Unique      | Official policy contract identification number     | "LIC1234567"                         |
| premiumAmount    | Number    | —           | Recurring premium installment amount               | 12000                                |
| premiumFrequency | String    | —           | Frequency: Monthly, Quarterly, Half-Yearly, Yearly | "Yearly"                             |
| premiumDueDay    | Number    | Conditional | Day of month (1 to 31) when premium payment is due | 5                                    |
| renewalDate      | Date      | Conditional | Next scheduled premium renewal due date            | 2026-09-05                           |
| coverageAmount   | Number    | —           | Total sum assured / maximum coverage limit         | 1000000                              |
| startDate        | Date      | —           | Policy coverage commencement date                  | 2026-01-01                           |
| maturityDate     | Date      | Conditional | Policy maturity date (for endowment policies)      | 2046-01-01                           |
| status           | String    | —           | Standing: Active, Expired, Matured, Closed         | "Active"                             |
| createdAt        | Date      | —           | Policy document creation timestamp                 | 2026-08-08                           |
| updatedAt        | Date      | —           | Policy document modification timestamp             | 2026-08-08                           |

## 7. Liability

Collection: liabilities

Tracks user debt obligations such as Home Loans, Vehicle Loans, Personal Loans, and Credit Cards, with EMI schedules and principal amortization.

| Field Name             | Data Type | Key                | Description                                                    | Example                              |
|------------------------|-----------|--------------------|----------------------------------------------------------------|--------------------------------------|
| <b>_id</b>             | ObjectId  | <b>PK</b>          | Liability debt identifier                                      | ObjectId("669a67890e1f234567890123") |
| <b>user</b>            | ObjectId  | <b>FK</b>          | Reference to debtor User document                              | ObjectId("65b8f1d2e13a4c001f8d901a") |
| <b>type</b>            | String    | —                  | Debt type: Home Loan, Vehicle Loan, Personal Loan, Credit Card | "Home Loan"                          |
| <b>name</b>            | String    | —                  | Title or description of the loan liability                     | "HDFC Home Loan"                     |
| <b>lender</b>          | String    | —                  | Creditor lending bank or financial institution                 | "HDFC Bank"                          |
| <b>principalAmount</b> | Number    | —                  | Original sanctioned loan amount borrowed                       | 1000000                              |
| <b>remainingAmount</b> | Number    | —                  | Current outstanding principal balance remaining                | 750000                               |
| <b>monthlyEMI</b>      | Number    | —                  | Regular Equated Monthly Installment (EMI) amount               | 10000                                |
| <b>interestRate</b>    | Number    | —                  | Annual percentage interest rate (p.a.)                         | 8.5                                  |
| <b>interestType</b>    | String    | <b>Conditional</b> | Interest structure (e.g. Fixed, Reducing)                      | "Reducing"                           |
| <b>dueDay</b>          | Number    | <b>Conditional</b> | Day of month (1 to 31) when EMI debits occur                   | 5                                    |
| <b>startDate</b>       | Date      | —                  | Loan disbursement origination date                             | 2025-01-01                           |
| <b>tenure</b>          | Number    | <b>Conditional</b> | Sanctioned repayment tenure in months                          | 120                                  |
| <b>nextDueDate</b>     | Date      | <b>Conditional</b> | Upcoming calendar date for next scheduled EMI                  | 2026-09-05                           |
| <b>status</b>          | String    | —                  | Servicing standing: Active, Paused, Overdue, Closed            | "Active"                             |
| <b>createdAt</b>       | Date      | —                  | Document creation timestamp                                    | 2026-08-08                           |
| <b>updatedAt</b>       | Date      | —                  | Document modification timestamp                                | 2026-08-08                           |

## 8. FinancialReminder

Collection: reminders

User financial calendar alert schedule for upcoming EMIs, SIPs, premium payments, and saving goal targets.

| Field Name            | Data Type         | Key       | Description                                                      | Example                              |
|-----------------------|-------------------|-----------|------------------------------------------------------------------|--------------------------------------|
| <b>_id</b>            | ObjectId          | <b>PK</b> | Unique reminder record identifier                                | ObjectId("66ab78901f2a345678901234") |
| <b>userId</b>         | ObjectId          | <b>FK</b> | Reference to recipient User document                             | ObjectId("65b8f1d2e13a4c001f8d901a") |
| <b>sourceType</b>     | String            | —         | Module related to the reminder: Liability, Investment, Insurance | "Liability"                          |
| <b>sourceId</b>       | ObjectId          | <b>FK</b> | Reference to related underlying financial document               | ObjectId("669a67890e1f234567890123") |
| <b>itemName</b>       | String            | —         | Title or description of the obligation item                      | "Home Loan EMI"                      |
| <b>dueDate</b>        | Date              | —         | Due date of the financial commitment                             | 2026-09-05                           |
| <b>notifyBefore</b>   | Array<br><Number> | —         | Days prior to due date for reminder notification                 | [5, 1, 0]                            |
| <b>channels.inApp</b> | Boolean           | —         | In-app alert delivery enabled                                    | true                                 |
| <b>channels.email</b> | Boolean           | —         | Email alert delivery enabled                                     | true                                 |

| Field Name | Data Type | Key | Description                                   | Example    |
|------------|-----------|-----|-----------------------------------------------|------------|
| amount     | Number    | —   | Monetary sum due for the commitment           | 10000      |
| status     | String    | —   | Reminder status: Scheduled, Sent, Completed   | "Active"   |
| read       | Boolean   | —   | Flag tracking whether user acknowledged alert | false      |
| createdAt  | Date      | —   | Reminder creation timestamp                   | 2026-08-08 |
| updatedAt  | Date      | —   | Last update timestamp                         | 2026-08-08 |

9. AIAdviceHistory

Collection: aisuggestions

Historical record of Gemini AI financial advisory recommendations, financial health assessments, and prioritized roadmap steps.

| Field Name      | Data Type      | Key | Description                                          | Example                                        |
|-----------------|----------------|-----|------------------------------------------------------|------------------------------------------------|
| _id             | ObjectId       | PK  | AI suggestion record identifier                      | ObjectId("66de01234c5d678901234567")           |
| user            | ObjectId       | FK  | User whose financial portfolio was analyzed          | ObjectId("65b8f1d2e13a4c001f8d901a")           |
| title           | String         | —   | Headline summary of the advisory recommendation      | "Accelerate Home Loan Prepayment"              |
| summary         | String         | —   | Executive summary synthesis of user financial health | "Prepaying Rs. 5000 saves 3.2 years of tenure" |
| category        | String         | —   | Strategic focus area: Savings, Debt, Investment      | "Debt Reduction"                               |
| overallHealth   | String         | —   | Holistic diagnostic score: Good, Excellent, Moderate | "Good"                                         |
| recommendations | Array <Subdoc> | —   | Embedded actionable recommendations and rationales   | [{ decision: "DEBT_FIRST", priority: "High" }] |
| actionSteps     | Array <Subdoc> | —   | Sequenced implementation steps generated by AI       | [{ step: 1, title: "Review liquid cash" }]     |
| modelUsed       | String         | —   | AI model engine utilized during inference            | "gemini-2.5-flash"                             |
| status          | String         | —   | Suggestion state: active, archived                   | "active"                                       |
| createdAt       | Date           | —   | Record creation timestamp                            | 2026-08-08                                     |
| updatedAt       | Date           | —   | Last update timestamp                                | 2026-08-08                                     |